

The Coefficient 8π and the Status of Issue 740

A derived coefficient on the OPH side and the withdrawal of a statement in Doc.
374

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The Coefficient 8π and the Status of Issue 740

Abstract. Doc. 374 carried the identification $E_\star = E_P$ as a reading **[S]** and concluded from it that #740 is closed under that reading. On the OPH side the coefficient has meanwhile been derived: from two observers seeing different stress it follows that $\kappa = bL^2/q$, so with the modular coefficient $b = 2\pi$ and the quarter area law $q = 1/4$, $\kappa = 8\pi L^2$ and $G = L^2$ **[Q]**. Hence $E_\star = \hbar c/L$ is the non-reduced Planck energy **[B]**. The further statement in Doc. 374 is withdrawn: OPH still carries the length identity, the clock and a matching factor as open, and eight of the fifteen dependencies of #740 are not closed there. Both frameworks have a matching factor at the same place: OPH 0.16 % between source ν and Fermi ν , FFGFT 1.10 % as the bare residual of the ladder (Doc. 375).

List of symbols

- [B]** algebraically proven
- [K]** numerically confirmed
- [S]** structurally plausible, derivation open
- [Q]** taken from an external source, cited

Symbol	Meaning
κ	coupling in Einstein's equation, $\kappa = 8\pi G/c^4$
b, q	modular coefficient and prefactor of the area law $S = A/(4L^2)$
L	length in the area law
$E_\star$	OPH reference energy of the hierarchy
$E_P, \bar{E}_P$	Planck energy, non-reduced and reduced
m	matching factor between source and convention quantity
ξ	FFGFT scaling parameter (4/30000)

1 Occasion

Doc. 374 set up the bridge equations to OPH and carried the identification $E_\star = E_P$ as a reading. Its result section states that #740 is closed under that reading. Two things have changed since: the coefficient has been derived on the OPH side, and OPH has set out the status of #740 in more detail. This document records both and corrects the statement that went too far.

2 The theorem [Q]

Two observers read the same stress and geometry configuration at the same metric norm. The tensor relation is $F_i = \kappa T_i + \varphi n_i$ with unknown κ and a common offset φ . On the entropy side there are a bulk variation $b K_i T_i$, an area variation $-K_i(F_i - \psi n_i)$ and stationarity $\Delta S_i + (q/L^2) \Delta A_i = 0$. If the two observers see different stress, $T_1 \neq T_2$, then, with nothing assumed about the offsets,

$$\kappa = \frac{bL^2}{q}. \quad (1)$$

With $b = 2\pi$ from the cap modular flow and $q = 1/4$ from the area law this gives $\kappa = 8\pi L^2$ [K], and from $\kappa = 8\pi G$ it follows that $G = L^2$.

3 Consequence for the reference energy

With $L = \sqrt{G}$ (in units $\hbar = c = 1$),

$$E_\star = \frac{\hbar c}{L} = \sqrt{\frac{\hbar c^5}{G}} = E_P, \quad (2)$$

the non-reduced Planck energy [B]. The reduced reading is off by $\sqrt{8\pi} = 5.013$ and would require a different normalisation of the area law; it is therefore not a choice of units but a different physical assumption.

The reading from Doc. 374 is thus no longer posited on the OPH side but derived. The prefactor 8π stays where it always was, in Einstein's equation; the quarter in the area law selects G .

4 Correction to Doc. 374

Doc. 374 concludes that #740 is closed under the reading $E_\star = E_P$. This is withdrawn. OPH carries #740 as the whole common-world join and names eight of fifteen dependencies as open [Q]. Open there are in particular:

1. The length in the area law and the length in the pixel are so far the same by declaration; deriving them as one length on one source family is planned.
2. The tick. $\hbar c/\ell_P$ is an energy; a clock requires an evolving state, a gap and a readout, plus the map to a laboratory transition. On the FFGFT side $\tilde{T} \cdot m = 1$ supplies the tick directly (Doc. 374, B3); on the OPH side it is work in progress.
3. The matching factor $m = v_F/v_{\text{source}} = 0.9984$, that is, the 0.16 %. It is to be computed from the same matter action.

What holds is therefore: the coefficient is derived, the bridge equations B1–B4 of Doc. 374 stand unchanged, and the absolute scale follows through the FFGFT chain. #740 itself remains open.

5 Matching factors in both frameworks

Both frameworks have an open factor at the same place, with different origins:

Framework	Factor	Size	Origin
OPH [Q]	$v_F/v_{\text{source}} = 0.9984$	0.16 %	source v vs. Fermi v , computation planned
FFGFT	$v_F/v(\xi) = 0.98912$	1.10 %	bare residual of the lepton ladder (Docs. 352, 375)

The FFGFT side knows the origin of its factor and carries it corpus-wide as bare versus measured; the OPH side carries its factor as open. For a comparison table both factors belong beside the value, as does the route (Doc. 375).

Tool and method

Claude (Anthropic) was used as a tool for: writing and running the Python check script, drafting the $\text{\LaTeX}$ sources, and maintaining register and changelog. The tool computes and writes; the author decides what is computed and written, judges every result and determines what it means physically. The theorem and the statements about the status of #740 are taken from the communication of the OPH author and marked **[Q]**.

6 Result

1. $\kappa = bL^2/q$ with $b = 2\pi$ and $q = 1/4$ gives $\kappa = 8\pi L^2$ and $G = L^2$ **[K]** **[Q]**.
2. Hence $E_\star = \hbar c/L$ is the non-reduced Planck energy **[B]**; the reduced reading is off by $\sqrt{8\pi}$.
3. The statement in Doc. 374 that #740 is closed under the reading is withdrawn; length identity, tick and matching factor remain open on the OPH side **[Q]**.
4. Matching factors: OPH 0.16 %, FFGFT 1.10 % as the bare residual **[K]**.

Check script: 2/python/Dok376_Skripte/pruef_376_koeffizient.py (9/9)

Register (Doc. 190): R124 (withdrawal of the #740 statement in Doc. 374).

Bibliography

- [1] J. Pascher, *Doc. 352 — Lepton residuals* (2026).
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- [3] J. Pascher, *Doc. 375 — The hierarchy ν/E_P* (2026).
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